



ELF ENERGY
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Develop the Modern
Grid, Together



Notebook tools for facilitating the usage of Dynawo's library of dynamic models

Clara Fernández (AIA)

Jose Luis Marin (AIA) · Guiu Oms (AIA) · Marco Chiaramello (RTE) · Florentine Rosiere (RTE)

The Dynawo Modelica library and its _INIT models

Dynawo, developed by RTE, includes a Modelica library of power system models. These notebooks are about using that library – not necessarily from Dynawo.

Dynawo's Modelica library

distributed with Dynawo

dynamic models

generators, loads, lines, transformers, etc.

_INIT models

some models have several, others none

DYNAWO'S INITIALIZATION PHILOSOPHY

```

graph LR
    A["Operating point  
(P,Q,U,theta)"] --> B["local init  
(_INIT models)"]
    B --> C["global init"]
    C --> D["simulation"]
    
```

Each _INIT model is solved independently, from the load flow values at its terminal. Its results become parameters of the dynamic model, not additional equations. Dynawo then solves the whole connected system at $t = 0$.

INITIALIZATION IN OPENMODELICA

To OpenModelica, an _INIT model is just another model. Nothing marks it as the initialization of anything.

The dynamic model still expects start values as parameters, and they have to be consistent with the load flow.

The notebooks that we present here make that link: they solve the _INIT models and write the results into the Modelica case.

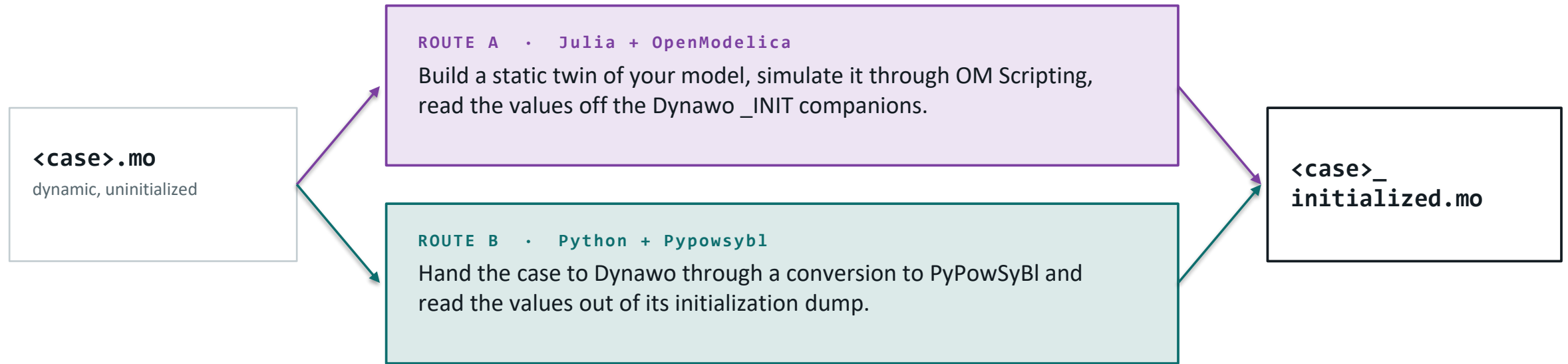
Rationale: since an _INIT model is a separate model, none of its equations enter the dynamic system.

Only the dynamic equations remain in the simulation, or in its linearization.



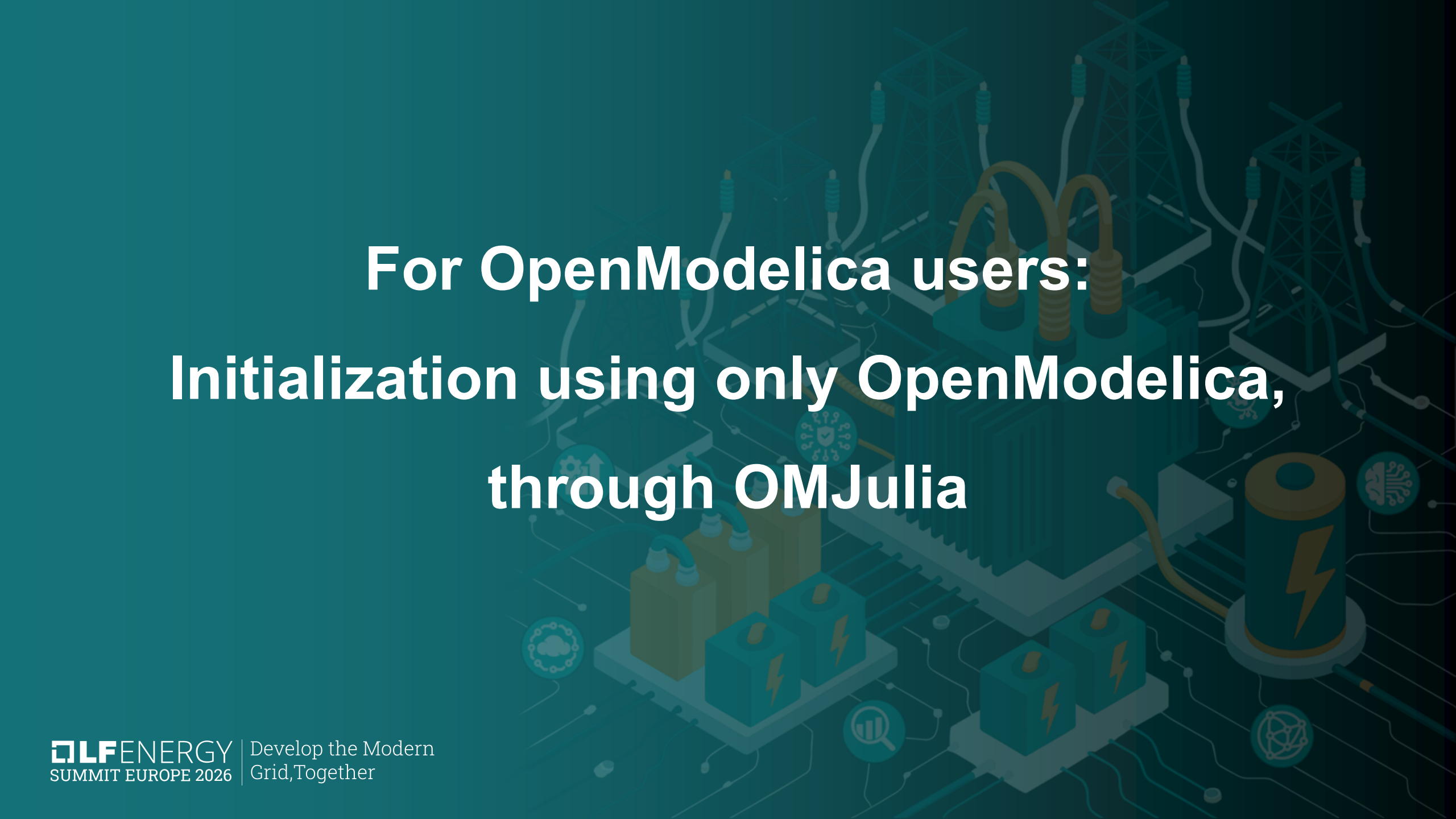
Initializing an OpenModelica case

Two routes. Same input, same output — the choice is which tools you would rather work with.



- Note the tools are **not** ready-to-use blackboxes covering all possible cases
- Rather, they are open, easy to understand code, easy to adapt to your particular modeling needs

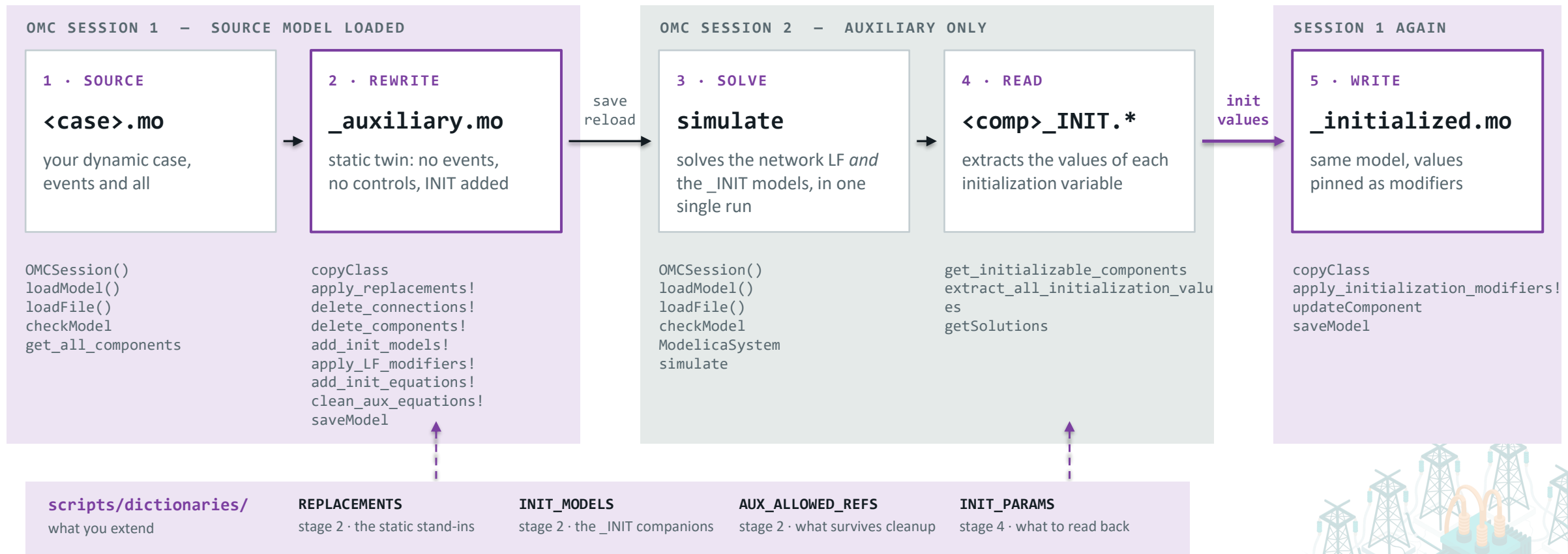




For OpenModelica users: Initialization using only OpenModelica, through OMJulia

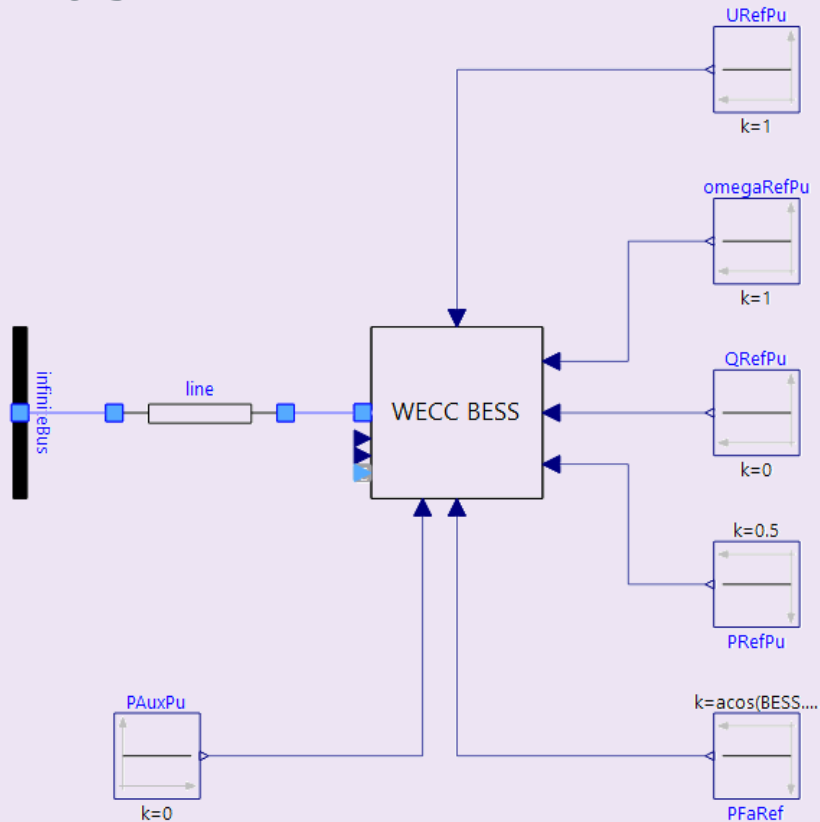
Initializing with OpenModelica alone

Build a static twin of the model, simulate it, and read the values off the _INIT companions.

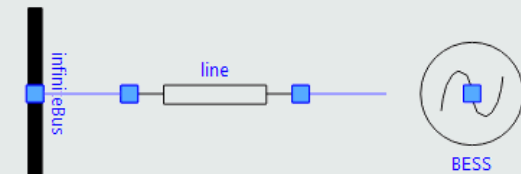


Example: Infinite Bus + Line + WECC BESS

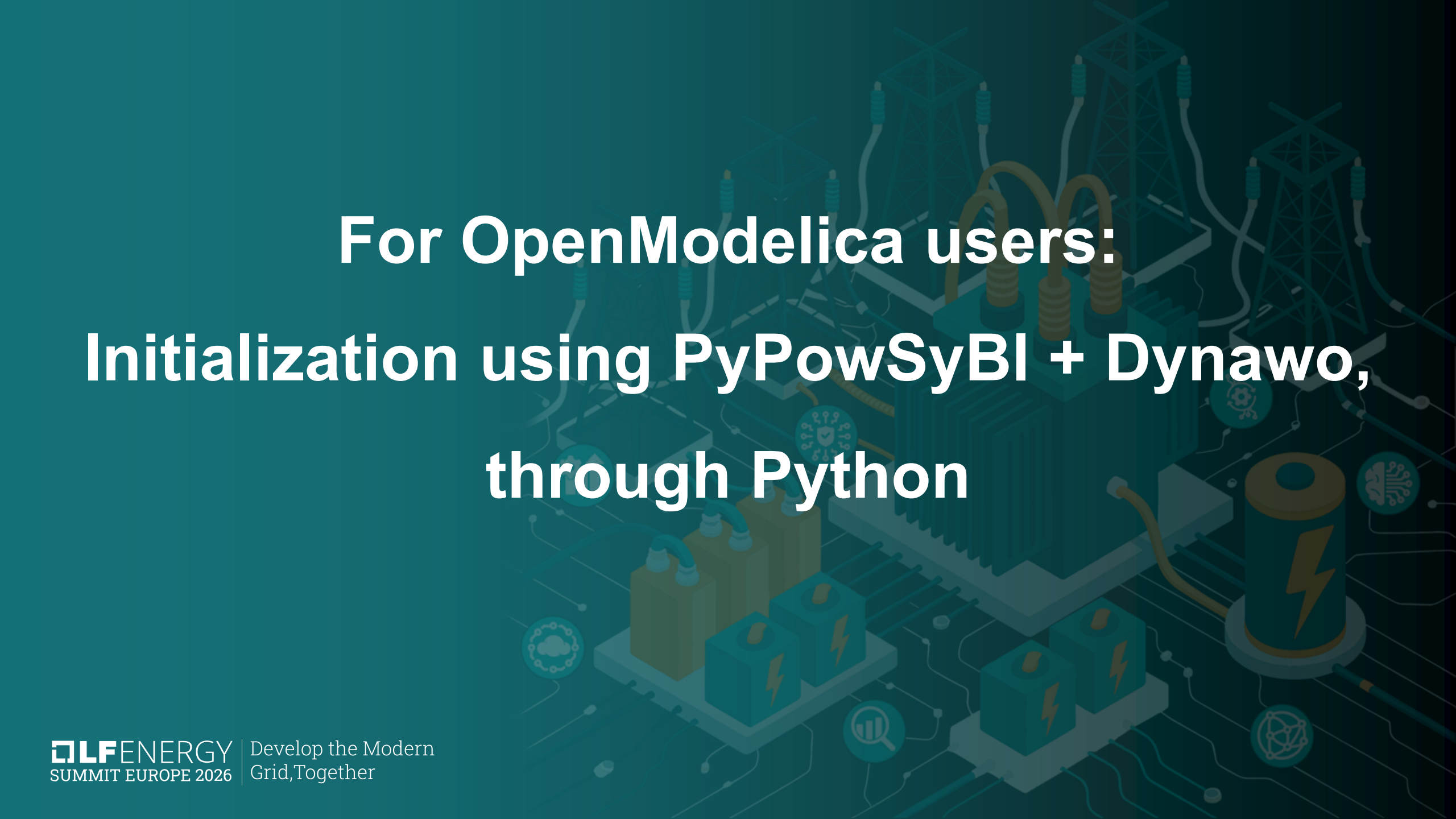
ORIGINAL CASE



AUXILIARY CASE



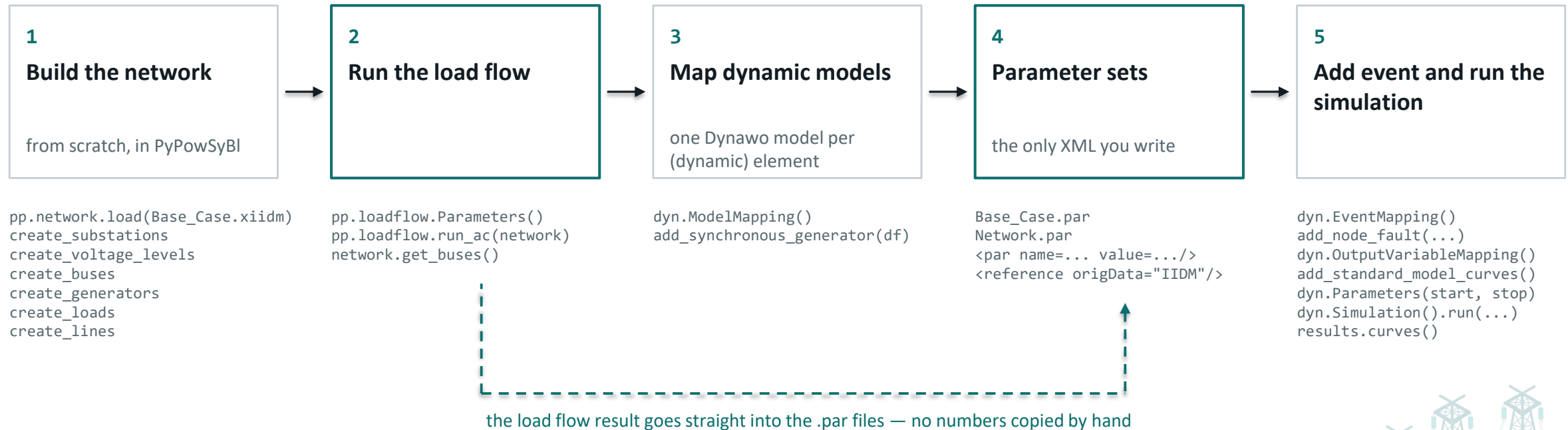
PV or PQ node, depending on the control mode of the original model



For OpenModelica users: Initialization using PyPowSyBI + Dynawo, through Python

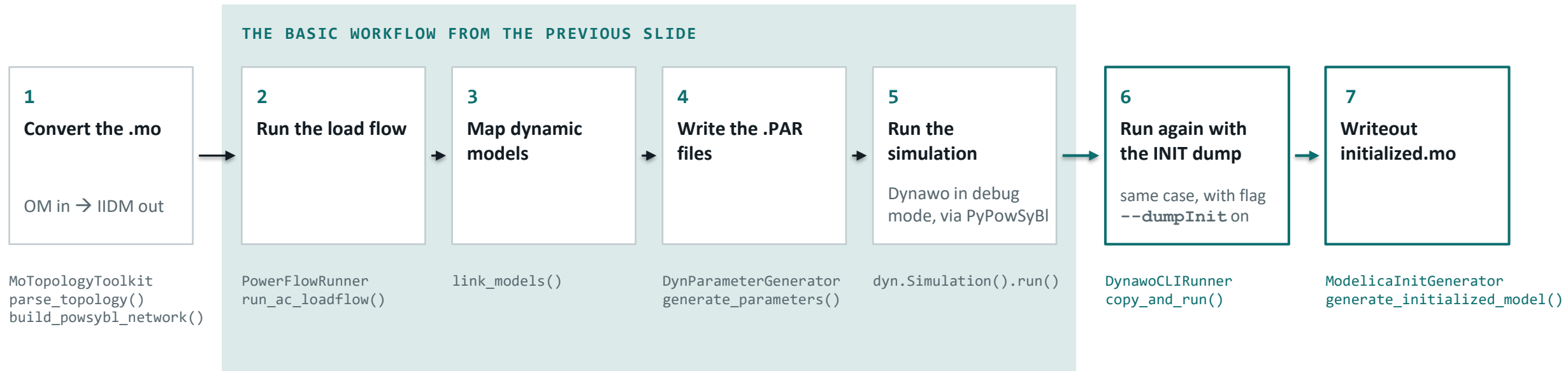
Running a case through PyPowSyBI and Dynawo

PyPowSyBI writes Dynawo's input files, so you never write a DYD by hand.



Initializing with PyPowSyBl + Dynawo

The same run, with the network converted from your OM case, and Dynawo's initialization values read back out.



Closing and Q&A